

Random Samples and Simple Probability Models

Probability as favourable outcomes over equally likely ones, scaling a sample up to a population, and judging whether a sample can be trusted

Directions:

- When every outcome is equally likely, $P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$. Give probabilities as fractions in lowest terms.
- To scale a sample up to a population, multiply the population size by the fraction the sample gives you.
- A sample is **random** when every member of the population has the same chance of being chosen. An estimate from a sample that is not random tells you about the group you asked, not about the population.
- Every quantity below is stated with its units. Answer in the same units.

1. A bag holds 12 counters, and 3 of them are red. One counter is drawn at random. What is the probability that it is red?

Answer: _____

2. A spinner is divided into 8 equal sections and 3 of them are shaded. The spinner is spun once. What is the probability that it lands on a shaded section?

Answer: _____

3. In a random sample of 50 students at a school, 18 walk to school. The school has 600 students in total. Estimate how many of the 600 walk to school.

Answer: _____

4. A box holds 12 counters and 5 of them are yellow. One counter is drawn at random. What is the probability that it is **not** yellow?

Answer: _____

5. A school of 800 students wants to know which sport is the favourite. A survey is given to the 25 members of the basketball team, and 22 of them choose basketball.

(a) Is this a random sample of the school? Answer on the line and explain in the work space: _____

(b) the number of the 800 students this sample would estimate choose basketball: _____

6. A spinner has 10 equal sections numbered 1 to 10. It is spun once.

(a) the probability of an even number: _____

(b) the probability of a number greater than 7: _____

7. A quality inspector takes a random sample of 80 light bulbs from a batch of 2000 and finds 6 faulty. Estimate the number of faulty bulbs in the whole batch.

Answer: _____

8. A club has 200 members and wants a sample of 20 of them.

Method 1: ask the first 20 members who arrive at the next meeting.

Method 2: number every member from 1 to 200 and let a computer pick 20 numbers at random.

(a) Which method gives a random sample? Write the method number on the line and explain in the work space: _____

(b) With that method, the probability that one named member is chosen: _____

9. A jar holds 20 marbles: 4 red, 6 blue and 10 white. One marble is drawn at random, its colour recorded, and it is put back.

- (a) the probability of drawing red: _____
- (b) the probability of drawing red or blue: _____
- (c) the number of reds you would expect in 200 draws: _____

10. Biologists caught 300 fish in a lake, tagged them and released them. Later they caught a random sample of 60 fish and found 9 of them tagged.

- (a) the fraction of the sample that was tagged: _____
- (b) the estimated number of fish in the lake: _____
- (c) Name one assumption this estimate depends on, and say what would happen to the estimate if tagged fish were easier to catch than untagged ones. Write this in the work space.